

innovate

achieve

lead



BITS Pilani
Pilani Campus

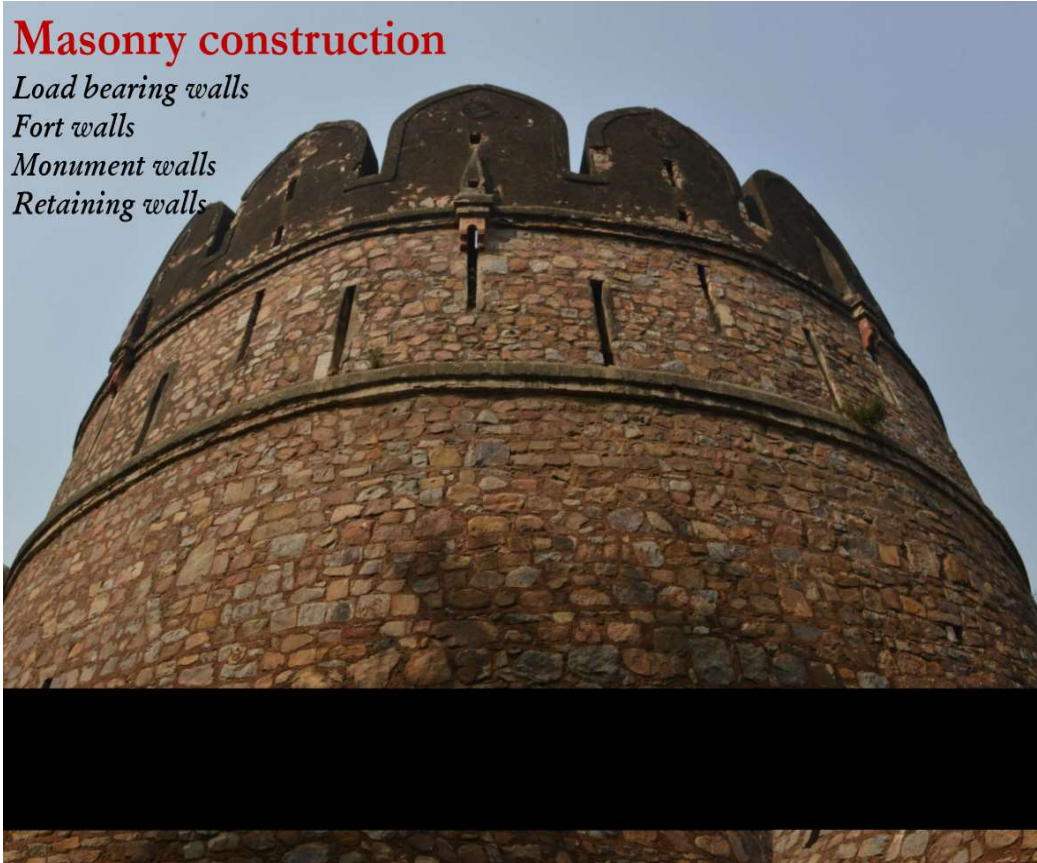
Lime

Lime in construction



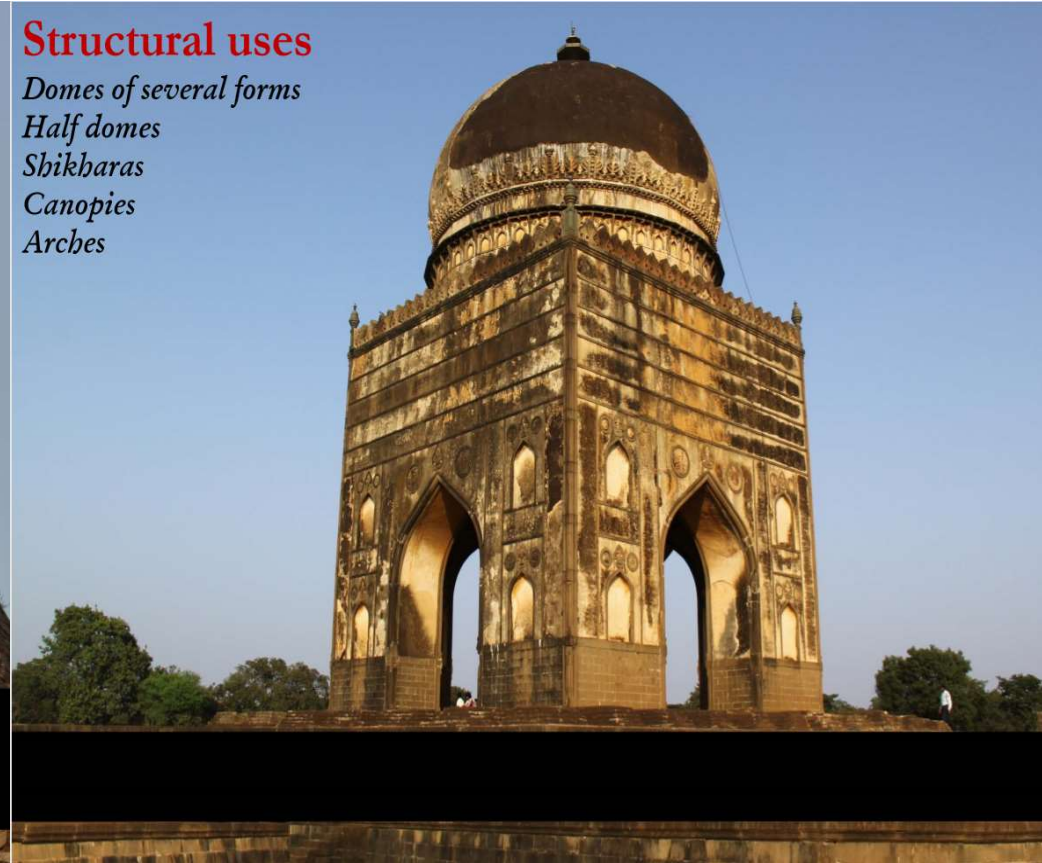
Masonry construction

Load bearing walls
Fort walls
Monument walls
Retaining walls



Structural uses

Domes of several forms
Half domes
Shikharas
Canopies
Arches



Lime in construction



Water structures- *Baolies, wells, water channels, fountains*



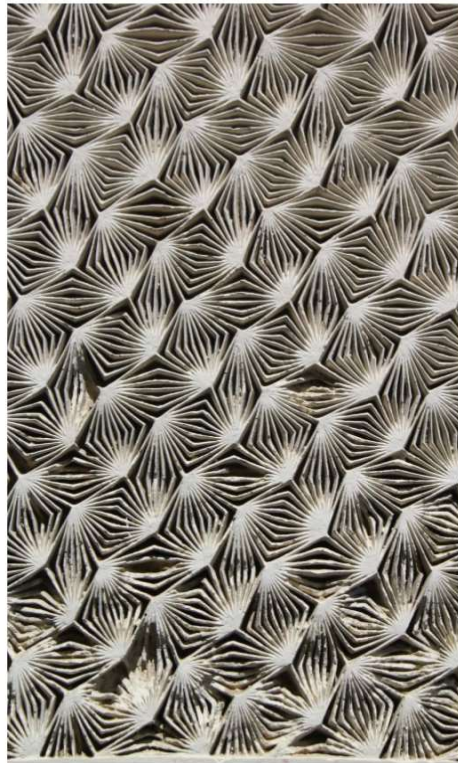
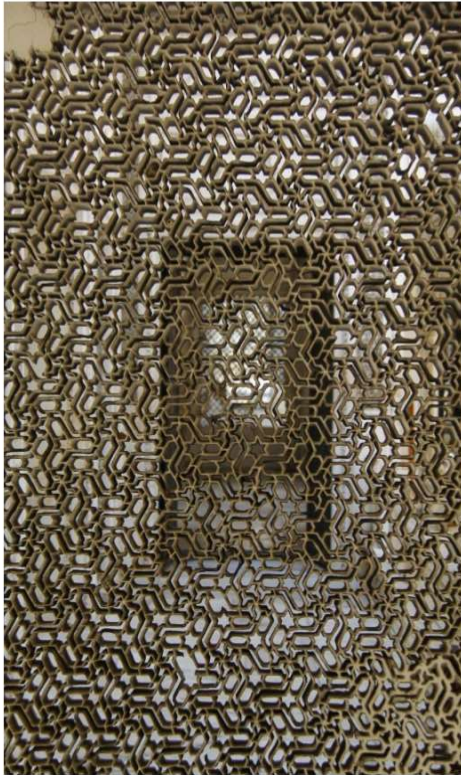
Plastering- *plain plastering, decorative plastering, incised plaster works*



Lime in construction



Jali works- *intricate patterns , geometrical profiles*



Decorative works- *stucco works, intricate patterns*



Introduction



- Lime was the chief cementitious material until the invention of Portland cement (Palaces, temples, forts, bridges)
- Cheap alternative to OPC at places where lime is locally available and if there is shortage of OPC
- Lime is good alternative to cement for
 - Lean concrete ($f_{ck} < 15$ MPa)
 - Plasters and mortars where the strength requirements are often low (< 3 MPa)

Introduction



- Raw material of lime is **calcium carbonate** which is obtained by calcination of limestone (Tufa, limestone boulders, kankars)
- Also obtained by calcination of shell, coral, chalk, other calcareous materials
- Lime is obtained by burning of limestone at about 800°C
$$\text{CaCO}_3 + 42.52 \text{ kcal} \text{ -----} \rightarrow \text{CaO} + \text{CO}_2$$
- Quick lime is unstable and reverts back to CaCO_3 . To prevent this **slaking of lime** is done i.e sprinkling of water

Varieties of lime



Pure limestone is also called calcite and that containing magnesite is called **dolomitic limestone**.

- **Stone lime** : Almost pure lime. Obtained by calcination of limestone. Has hydraulic properties.
- **Kankar lime** : Impure lime. Obtained from calcination of kankars. Used for making hydraulic lime.
- **Shell lime** : Very pure lime obtained by calcination of shells of sea animals and corals. Lime punning, white wash, color wash.
- **Magnesian lime** : Manufactured from dolomite. Contains magnesia more than 5 %. Mortar, plaster.

Definitions



- **Quick lime:** White oxide of calcium. Presence of magnesium oxide gives it a brownish or grayish tinge. Also called *caustic lime*. Specific gravity = 3.40
- **Fat lime :** High calcium oxide component, sets and hardens by absorption of CO_2 from atmosphere. Burning pure limestone, marble, white chalk, sea shell, coral.
- **Hydraulic lime :** Contains small quantities of silica, alumina, iron oxide, alongwith CaO. It has property to set and harden under water.
- **Hydrated Lime:** When quick lime is finely crushed, slaked with a minimum amount of water and screened or ground to form a fine powder.
- **Lump lime :** Quick lime coming out of the kilns.
- **Milk Lime :** Thin pourable solution of slaked lime in water

Characteristics of lime



- Lime possesses **good plasticity** and is easy to work with
- It **stiffens easily** and is **resistant to moisture**
- The **excellent cementitious properties** make it most suitable for masonry work
- The shrinkage on drying is small because of its high water retention
- **Uses:**
 - As mortar for laying masonry and for plastering (slaked lime)
 - **Lime punning** about 3 mm thick **shell lime layer** is used to improve the plastered surfaces and to give a shining appearance.
 - Lime bricks, artificial stones, paints, glass, soil stabilization, flux

Manufacture of lime



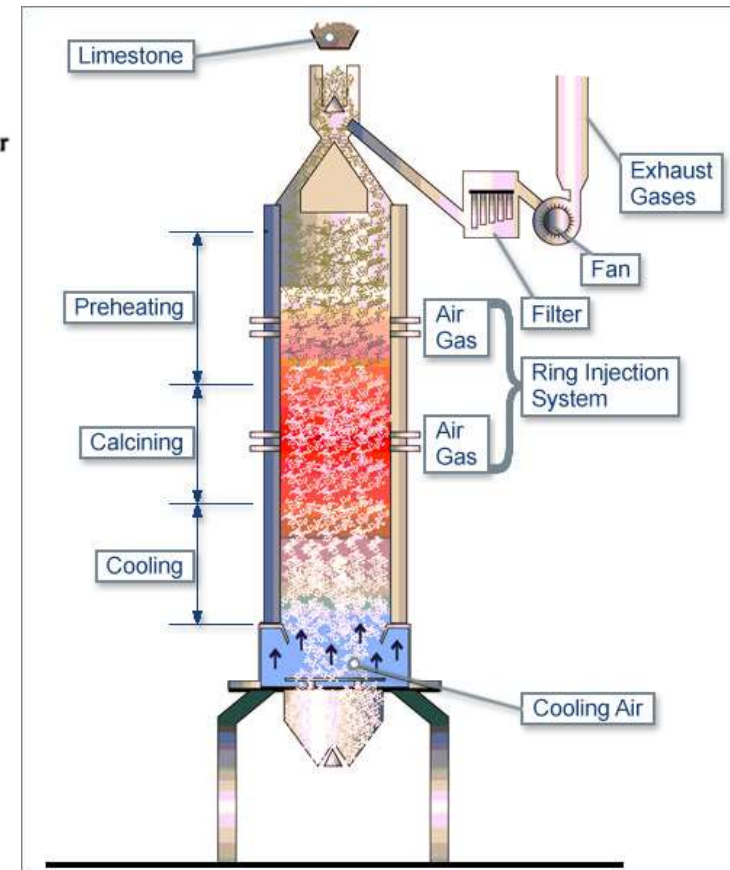
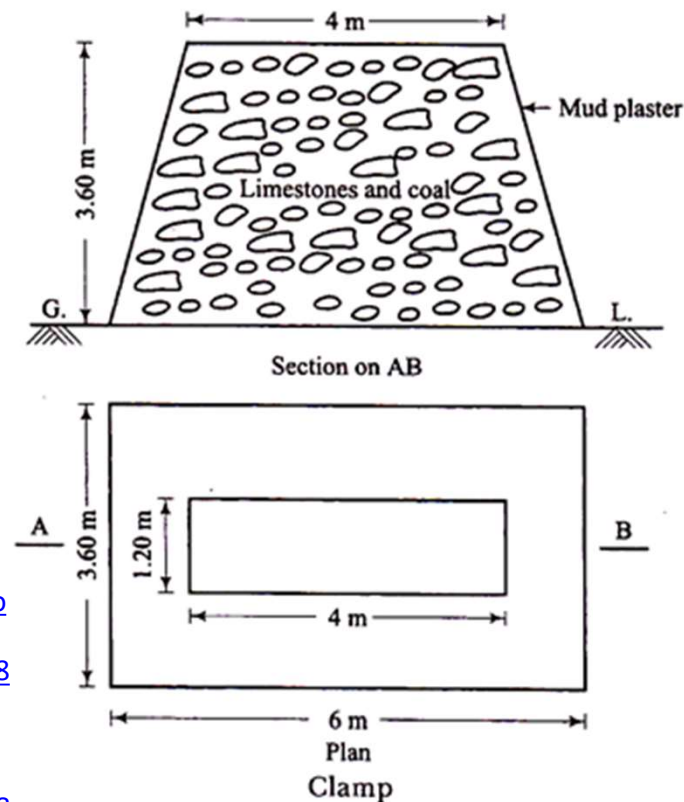
- Preparation, calcination, slaking
- Calcination = burning of limestone:
 - Clamps,
 - intermittent kilns,
 - continuous kilns

<https://www.youtube.com/watch?v=9LDG9cnGIDo>

<https://www.youtube.com/watch?v=VqSwHLpOd88>

<https://www.youtube.com/watch?v=luHyJbqQ5Tk>

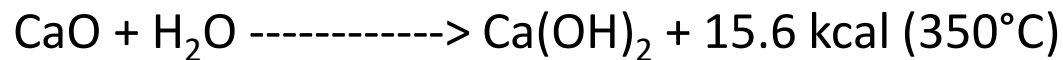
https://www.youtube.com/watch?v=L_RD5jvXP68



Slaking of lime



- Quick lime when sprinkled with water, slakes within 10 minutes and becomes powder.
- White washing, plastering, mortars, lime putty



- Theoretically water required 32% of the weight of the CaO. However it may be 2 or 3 times higher based on the composition, degree of burning and slaking methods.
- Fat lime slakes in 2-3 hours, whereas hydraulic lime slakes in 12-48 hours.
- Lime if not slaked in time may react with carbonic acid in air and form calcium carbonate.

Products of lime slaking

Lime putty

Bottommost layer, thick, cheesy consistency

Milk of lime

Water above the lime putty, white with suspended calcium hydroxide

Lime water

Topmost layer, clean water with suspended particles of calcium hydroxide



Tank slaking

Classification



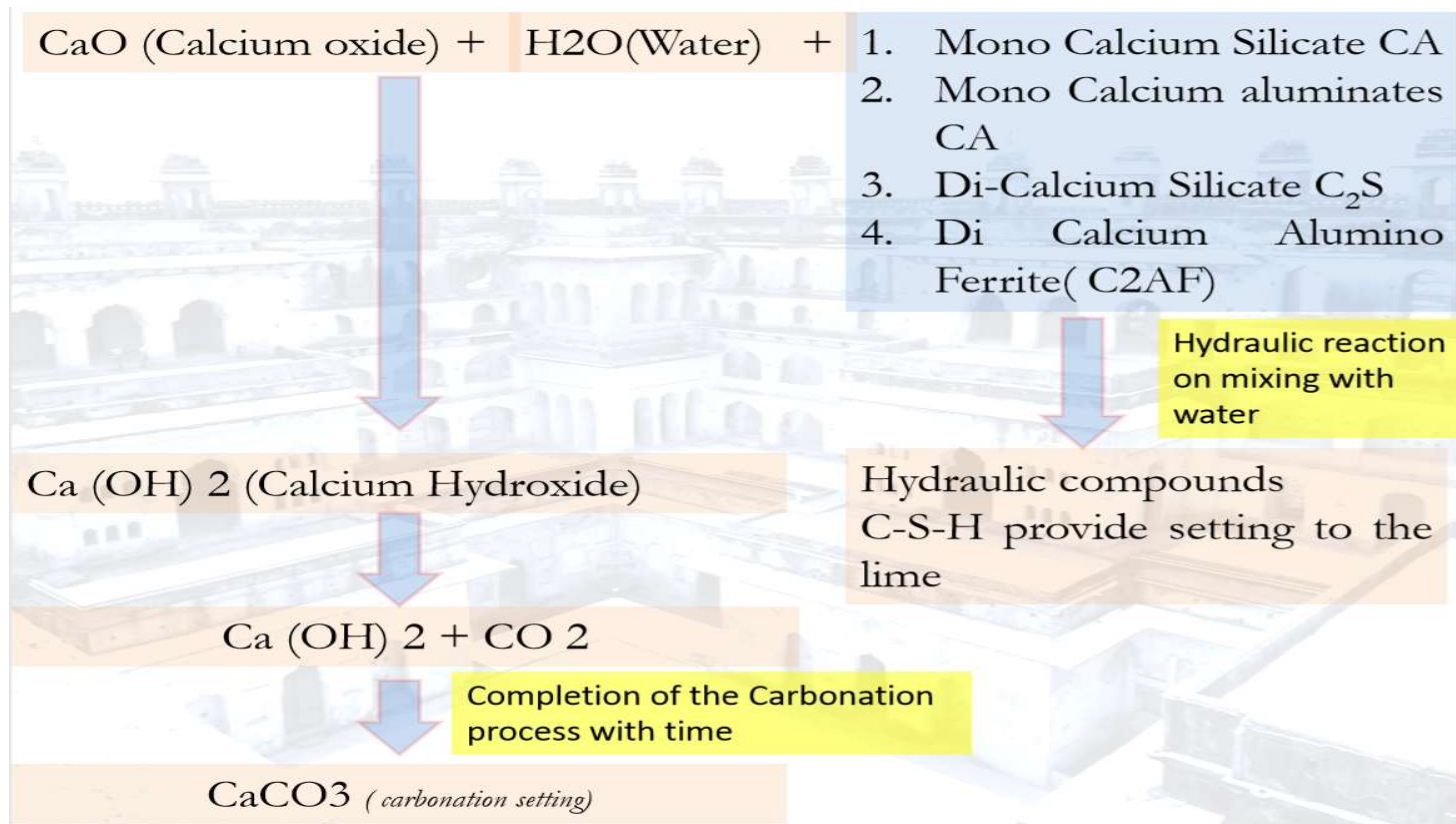
- Lean or poor lime
- Hydraulic lime
- Pure, rich or fat lime

Lean or poor lime



- More than 30% clay
- **Characteristics:**
 - Setting and hardening is very slow
 - Poor binding property
 - The colour varies from yellow to grey (dull color)
 - Recommended for less important structures

Hydraulic lime



- Sets by chemically combining with water
- 5-30% clay
- Based on hydraulicity, feeble (5-10%) moderately (10-20%), eminently hydraulic (20-30%) lime
- Mortar for plastering work

Pure, rich or fat lime



- About 95% CaO, less than 5% impurities
- **Characteristics:**
 - Slakes vigorously and the volume becomes 2-3 times
 - Colour perfectly white, high degree of plasticity
 - It sets slowly in contact with air and hence is not suitable for thick walls or in wet climate
 - If kept under water a fat lime paste does not lose its high plasticity and consequently does not set and harden
 - Sp. Gr of pure lime is 3.4
 - White washing, mortar for brick work and plastering, *surkhi* mortar for building foundations and thick masonry walls

Classification (IS:712)



Class A Eminently Hydraulic lime	• Colour is grey. • Contains 60-70% CaO and 25% clay • Slakes with Difficulty	• Sets and hardens readily under water with initial setting time 2 hrs and final setting time 48 hrs • Used for making mortar and concrete for structures
Class B Semi Hydraulic lime	• Colour is grey • Contains 70% CaO and 15% clay • Slakes and sets slow rate taking about a week	• Used for masonry mortars, flooring and for concrete in ordinary constructions and plaster undercoat
Class C Fat lime	• Colour is white • Slakes vigorously and increase 2-3 times in Vol • Contains about 93% CaO and 5-7% clay	• Used for finishing coat in plastering and white washing, and with pozzolona in mortars
Class D- Magnesium/dolo mite lime	• Colour is white • Contains about 85% of Ca and Mg oxides • Slakes promptly	• Sets slowly • Used for finishing coat in plastering and white washing
Class E – Kankar lime	• Colour is grey • Contains 20% CaO, 5% MgO and remaining impurities	• Slakes and sets slowly • Used in masonry mortars, finishing and white washing
Class F – Siliceous dolomite lime	• Used for undercoat and finishing coat of plaster	

Lime putty



- It is obtained by adding hydrated lime to water, stirring to the consistency of a thick cream and, allowing it to stand and mature for a period of about 16 hours in the case of non- hydraulic lime before using it for making mortar.
- The putty so obtained should be protected from drying out

https://www.google.com/search?q=lime+putty+meaning&sca_esv=579056776&tbm=vid&source=lnms&sa=X&ved=2ahUKewiK94yn_aaCAxVETmwGHRZ6CyAQ_AUoA3oECAEQBQ&biw=1280&bih=585&dpr=1.5#fpstate=ive&vld=cid:637b3022,vid:LBI8fdL75Mw,st:0

Field tests



- ***Test for Physical Properties***

- Hydraulic limes are bluish grey, brown or dark coloured. They have a clayey taste and give out an earthy smell.
- White colour is an indication of pure variety of limestone.
- Shining particles on the surface indicate the presence of free salt

- ***Workability test***

- A handful of mortar(1:3) is thrown on the wall, area covered is used to roughly determine the workability

Field tests



- ***Impurity test***

A solution of lime mixed with water is decanted. The residue is dried for 8 hours in hot sun and is weighed. Less than 10 % is good, 10-20% is fat and more than 20% is poor.

- ***Heat test***

Limestone is burnt for 4 hours. If the residue is lumpy then it is unburnt limestone. And porous indicates quicklime.

- ***Ball test***

Physical tests



- ***Fineness test***

- Sieves are arranged with coarser on top and finer at bottom.
- Sieving is done
- 100 g of hydrated lime is placed
- The residue on each sieve is dried and measured
- Percentage of each sieve is calculated

Workability test and setting time



- <https://www.youtube.com/watch?v=Wzg3DPvtJ-I>
- Setting time is determined using Vicats apparatus
 - https://www.youtube.com/watch?v=Y4vzX9_YCYc

Soundness test



- Cement, hydrated lime and standard sand are mixed in 1:3:12
- 12% by mass of water is added to the dry mix and three moulds are prepared
- Moulds are covered with non porous plates and small weights
- The apparatus is kept damp for 48 hours and then subjected to steam for 3 hours
- The distance between the pointers should not be more than 10 mm.

<https://www.youtube.com/watch?v=je5ztHs9tII>

Compressive strength test



- Twelve cubes with sides 50 mm are prepared from standard lime sand mortar (1:3) and kept for 72 hrs at 90% humidity
- Next they are kept in the air for 4 days
- Six cubes are cured in water for 7 days and tested in compression testing machine (14 d strength), remaining cubes are cured for 21 days (28 d strength).
- The crushing load is divided by area of the cube to get the compressive strength of mortar

Storage of lime



Lime has to be protected from Moisture

- It should be stored in **close store rooms** to avoid air slaking
- When delivered in hydrated form, it should be stored **under cover and off ground**. It should be used quickly on site
- Lime putty is stored without any deterioration for many weeks and improves by keeping. In case of semi hydraulic lime, the putty must not be stored for more than 3 days of its preparation

Precaution in handling lime



- Slaking of lime produces enormous heat and dust.
- The workers must be provided with goggles and respirators since high heat and dust is generated during slaking
- Skin protecting cream must be used by the workers as protection against skin burn

Lime vs Cement



Property	Lime	Cement
Colour	White or greyish white	Dark grey or grayish brown
Slaking	Slaking takes place on adding water	Hardening takes place on adding on water
Setting	It sets slowly by taking CO ₂ from air or by reacting with water	It sets rapidly by reacting with water
Hardening	Slow	Rapid
Compressive strength	Less	High
Cost	Cheap	Costly
Use	Suitable for ordinary construction works	Suitable for all construction works



Thank you